

Team Contributions & Project Roles

Team Member	Module / Role	Key Engineering Contributions
Amr Khaled Mohamed Ali	System Integration & Architecture Lead	- Designed top-level wrappers (<code>perceptron_top</code> , <code>perceptron_layer</code>). - Implemented spatial parallelism via <code>generate</code> loops & dynamic indexing (<code>+:</code>). - Designed pipeline registers & sign-extension logic for timing closure.
Omar Sherif Zaki	Verification & Testing Engineer	- Developed testbenches (<code>tb_perceptron_layer</code>) for system verification. - Managed stimulus generation, signal timing checks, and test scenario coverage.
Omar Mohamed Salem	MAC Unit Designer	- Implemented the Multiply-Accumulate (<code>mac_unit</code>) hardware core. - Integrated 2's complement arithmetic & signed accumulation logic.
Mariam Hossam Mohamed	Control Unit Engineer	- Designed the Finite State Machine (<code>fsm_controller</code>). - Managed system handshaking, synchronization signals (<code>mac_en</code> , <code>clear</code>), and state flow.
Roaa Sheriff Sayed	Data Post-Processing Engineer	- Designed the <code>Bias_Add_Unit</code> with bit-width sign expansion. - Implemented the non-linear <code>ReLU</code> activation function module.